

# SIXTH SEMESTER [B.TECH]

## STATISTICS, STATISTICAL MODELLING & DATA ANALYTICS

### UNIT - I

#### DATA VISUALIZATION

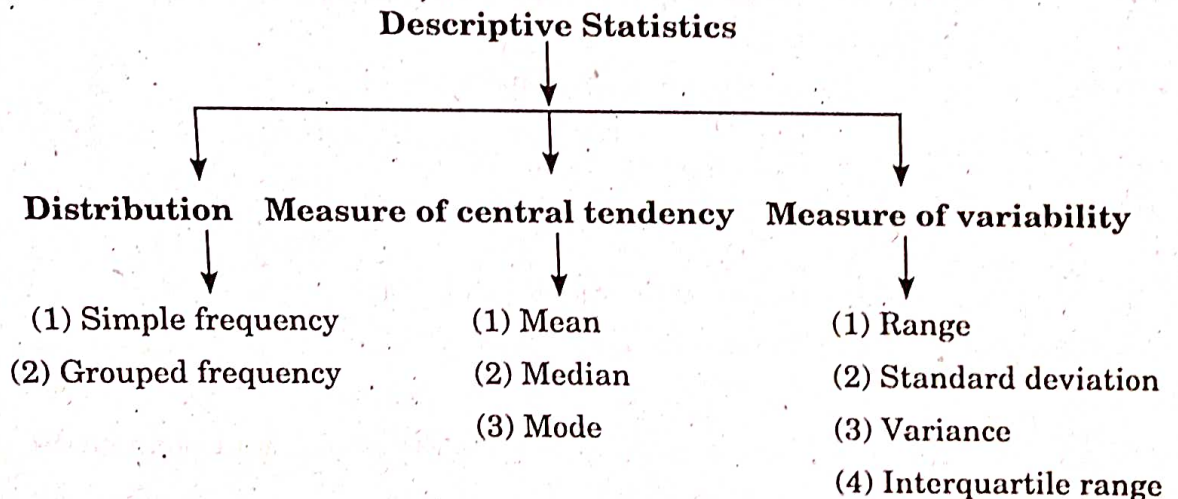
- (1) Bar chart - presents categorical data with rectangular bars
- (2) Histogram - vertical bar chart of frequencies
- (3) Frequency polygon - line graph of frequencies
- (4) Ogive - line graph of cumulative frequencies
- (5) Pie Chart - proportional representation for categories of a whole.
- (6) Stem and Leaf plot - sorting the data and graphing it at the same time.
- (7) Pareto chart - mix of a bar graph and a line graph and used for demonstrating the 80-20 principle
- (8) Scatter plot - shows relationship between two variables.

#### Descriptive Statistics

Descriptive statistics summarize and organize characteristics of a data set. A data set is a collection of responses or observations from a sample or entire population.

There are 3 main types of descriptive statistics:

1. The **distribution** concerns the frequency of each value.
2. The **central tendency** concerns the averages of the values.
3. The **variability** or **dispersion** concerns how spread out the values are.



**Mean** - average of given data

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where,  $x_i$  = observations,  $n$  = no. of observation

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

where,  $x_i$  = value of the observation or class mark

$f_i$  = frequency for  $i^{\text{th}}$  observation or class interval

**Median** - middle observation after arranging data into ascending or descending order.

**For raw data**

(i)  $n$  = no. of observations

If  $n$  is odd

$$\text{Median} = \left( \frac{n+1}{2} \right)^{\text{th}} \text{ term}$$

(ii) If  $n$  is even

$$\text{Median} = \frac{\left( \frac{n}{2} \right)^{\text{th}} \text{ term} + \left( \frac{n}{2} + 1 \right)^{\text{th}} \text{ term}}{2}$$

**For class interval data**

$$\text{Median} = l_0 + \left\{ \frac{\left( \frac{N}{2} \right) - C}{f_0} \right\} h$$

$l_0$  = lower limit of median class

$f_0$  = frequency of median class

$C$  = cumulative frequency before median class

$h$  = width of median class.

**Mode** - having highest frequency in the data

$$\text{Mode} = l_0 + \left( \frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1} \right) h$$

$f_{-1}$  = frequency before mode class

$f_1$  = frequency after mode class

$f_0$  = frequency of mode class

$h$  = width of mode class

$l_0$  = lower limit of mode class

**Range** - difference between the largest value ( $L$ ) & the smallest value ( $S$ )

$$\text{Range} = L - S$$

**Co-efficient of range** - a relative measure of range

$$\text{Co-efficient of range} = \frac{L - S}{L + S}$$

**Quartile deviation**

Lower quartile ( $Q_1$ )

Upper quartile = ( $Q_3$ )

$$\text{Inter quartile range} = Q_3 - Q_1$$

$$\text{Quartile deviation} = \frac{Q_3 - Q_1}{2}$$

$$\text{Co-efficient of quartile deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$



**For raw data**

Arrange the data in ascending order

$$Q_1 = \left( \frac{N+1}{4} \right)^{\text{th}} \text{ term}$$

$$Q_3 = \left[ \frac{3(N+1)}{4} \right]^{\text{th}} \text{ term}$$

**For class interval data**

$$Q_1 = l_1 + \frac{l_2 - l_1}{f} \left( \frac{N}{4} - cf \right)$$

$$Q_3 = l_1 + \frac{l_2 - l_1}{f} \left( \frac{3N}{4} - cf \right)$$

$l_1$  = lower limit of quartile class

$l_2$  = upper limit of quartile class

$f$  = frequency of quartile class

$cf$  = cumulative frequency before quartile class

**Standard Deviation** - measures the absolute dispersion

$$\sigma_x = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

**Variance** - square of S.D.

$$\text{Variance} = \sigma_x^2 = \frac{\sum (X - \bar{X})^2}{N}$$

$$\text{Co-efficient of variance} = \frac{\sigma_x}{\bar{X}} \times 100$$

**Probability Distribution**

(1) Discrete Probability Distribution

(2) Continuous Probability Distribution

**Discrete Probability Distribution**

A discrete probability distribution can always be represented in a tabular form as follows:

|      |       |       |     |       |
|------|-------|-------|-----|-------|
| X    | $x_1$ | $x_2$ | ... | $x_n$ |
| P(x) | $P_1$ | $P_2$ | ... | $P_n$ |

where  $\sum_{i=1}^n P_i = 1$  &  $P_i \geq 0 \forall i = 1, 2, \dots, n$

$$\text{Mean} = E(x) = \sum_{i=1}^n x_i P_i$$

$$\begin{aligned} \text{Variance} = \sigma^2 &= E(X - \mu)^2 = \sum_{i=1}^n P_i (x_i - \mu)^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

## Continuous Probability Distribution

Let  $X$  is a continuous random variable and  $f(x)$  is continuous function of  $x$  called probability density function then the curve  $y = f(x)$  is called probability density curve. The probability of continuous random variable in a given interval  $(a, b)$  is given by

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

The density function  $f(x)$  must satisfy

(i)  $f(x) > 0 \quad \forall x$

(ii)  $\int_{-\infty}^{\infty} f(x) dx = 1$

## Sampling Distribution

**Population** - Any collection of individuals under study is said to be population.

**Sample** - A part or small section selected from the population is called a sample.

**Parameters** - Statistical measurement such as mean, variance, etc. of the population are called parameters.

**Sampling** - The process of forming a sample consists of selecting a predetermined number of individuals from the parent universe is called sampling.

**Statistical Hypothesis** - It may be defined as a statement about one or more population.

**Null Hypothesis** - It is the hypothesis which is tested for possible rejection under the assumption that it is true. This hypothesis states that the true difference between the two values is zero. It is denoted by  $H_0$ .

**Alternative Hypothesis**: Any hypothesis which is complementary to the null hypothesis is called an alternate hypothesis denoted by  $H_1$ . The rejection of null hypothesis means the acceptance of alternate hypothesis.

**Critical Region** - The part of a sample space which amounts to rejection of null hypothesis  $H_0$  is called critical region.

**Test of significance** - Test used to ascertain whether the differences (between estimator and the parameter or between two estimators) are real or due to chance are called test of significance.

**Level of significance**: In testing a given hypothesis, the maximum probability with which we would be willing to risk an error is called level of significance of the test. In practice a level of significance 0.05 (5%) or 0.01 (1%) is usually taken.

## Large Sample Test (z-distribution)

**Null hypothesis**

$$\mu = \mu_0$$

$$\mu_1 = \mu_2$$

**Test statistic**

$$z = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$



$$p = p_0$$

$$z = \frac{p_1 - p}{\sqrt{\frac{pq}{n}}} = \frac{z - p}{\sqrt{npq}}$$

$$p_1 = p_2$$

$$z = \frac{p_1 - p_2}{\sqrt{pq \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

### Small sample Test (t-test)

#### Null hypothesis

$$\mu = \mu_0$$

#### Test statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}, \text{ where } s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$\mu_1 = \mu_2$$

$$t = \frac{\bar{x} - \bar{y}}{\frac{s}{\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}, \text{ where } s^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_j - \bar{y})^2}{n_1 + n_2 - 2}$$

#### Paired t-test

$$\mu_1 = \mu_2$$

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}, \text{ where } s_d^2 = \frac{1}{r-1} \sum (d_i - \bar{d})^2$$

### $\chi^2$ - distribution

#### Null hypothesis

Fit is good

#### Test statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

The two attributes are independent

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

### F-distribution

#### Null hypothesis

#### Test statistic

$$\sigma_1^2 = \sigma_2^2$$

$$F = \begin{cases} \frac{S_1^2}{S_2^2} & S_1^2 > S_2^2 \\ \frac{S_2^2}{S_1^2} & S_2^2 > S_1^2 \end{cases}$$

$$\text{where } S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}, S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

**Q.1.** The means of simple samples of 1000 & 2000 are 67.5 and 68.0 inches respectively. Can the samples be regarded as shown from the same population of S.D. 2.5 inches.

**Ans.** Let  $H_0$  : Samples are drawn from same population

We have  $n_1 = 1000$ ,  $n_2 = 2000$ ,  $\bar{x}_1 = 67.5$ ,  $\bar{x}_2 = 68.0$ ,  $\sigma = 2.5$

The Z-statistic is given by

$$\begin{aligned} Z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} \\ &= \frac{67.5 - 68.0}{\sqrt{(2.5)^2 \left( \frac{1}{1000} + \frac{1}{2000} \right)}} = -5.16 \end{aligned}$$

Since  $5.16 = |Z_{\text{cal}}| > Z_{\text{tab}} = 2.58$  at 1% level of significance. So, null hypothesis is rejected.

**Q.2.** The average number produced by two machines per day are 200 and 250 with S.D. 20 and 25 respectively on the basis of records of 25 days production. Can you regard both the machines equally efficient at 1% level of significance.

**Ans.** Let  $H_0$  : Both the machine are equally efficient

$n_1 = 25 = n_2$ ,  $\bar{x}_1 = 200$ ,  $\bar{x}_2 = 250$ ,  $\sigma_1 = 20$ ,  $\sigma_2 = 25$ .

$$\begin{aligned} S &= \sqrt{\frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2 - 2}} \\ &= \sqrt{\frac{25 \times 20^2 + 25 \times 25^2}{25 + 25 - 2}} = 23.105 \end{aligned}$$

t - statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\frac{S}{\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}} = -7.65$$

Since  $7.65 = |t_{\text{cal}}| > t_{\text{tab}} = 2.68$  at 1% level of significance for 48 degree of freedom, so null hypothesis is rejected.

**Q.3.** A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Are there figures commensurate with general examination result which is in the ratio of 4:3:2:1 for the various categories respectively.



**Ans.** Let  $H_0$  : Observed results are commensurate with general examination results

| Class  | Observed frequency ( $O_i$ ) | Expected frequency ( $E_i$ )    | $(O_i - E_i)^2$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|--------|------------------------------|---------------------------------|-----------------|-----------------------------|
| Failed | 220                          | $\frac{500 \times 4}{10} = 200$ | 400             | $\frac{400}{200}$           |
| Third  | 170                          | $\frac{500 \times 3}{10} = 150$ | 400             | $\frac{400}{150}$           |
| Second | 90                           | $\frac{500 \times 2}{10} = 100$ | 100             | $\frac{100}{100}$           |
| First  | 20                           | $\frac{500 \times 1}{10} = 50$  | 900             | $\frac{900}{500}$           |
|        | 500                          | 500                             |                 | $\frac{71}{3}$              |

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{71}{3} = 23.67$$

Since  $23.67 = \chi_{cal}^2 > \chi_{tab}^2 = 7.81$  at 5% level of significance for 3 degree of freedom.

So null hypothesis is rejected.

**Q.4.** Two samples are drawn from two normal population. From the data information test whether the two samples have the same variance at 5% level of significance.

|          |    |    |    |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|----|----|
| Sample 1 | 60 | 65 | 71 | 74 | 76 | 82 | 85 | 87 |    |    |
| Sample 2 | 61 | 66 | 67 | 85 | 78 | 63 | 85 | 86 | 88 | 91 |

**Ans.** Let  $H_0$  : The samples have the same variance

| Sample 1(x) | Sample 2(y) | $x - \bar{x}$ | $(x - \bar{x})^2$ | $y - \bar{y}$ | $(y - \bar{y})^2$ |
|-------------|-------------|---------------|-------------------|---------------|-------------------|
| 60          | 61          | -15           | 225               | -16           | 256               |
| 65          | 66          | -10           | 100               | -11           | 121               |
| 71          | 67          | -4            | 16                | -10           | 100               |
| 74          | 85          | -1            | 1                 | 8             | 64                |
| 76          | 78          | 1             | 1                 | 1             | 1                 |
| 82          | 63          | 7             | 49                | -14           | 196               |
| 85          | 85          | 10            | 100               | 8             | 64                |
| 87          | 86          | 12            | 144               | 9             | 81                |
|             | 88          |               |                   | 11            | 121               |
|             | 91          |               |                   | 14            | 196               |
| 600         | 770         |               | 636               |               | 1200              |

$$\bar{x} = \frac{\Sigma x}{n_1} = \frac{600}{8} = 75$$

$$\bar{y} = \frac{\Sigma y}{n_2} = \frac{770}{10} = 77$$

$$S^2_1 = \frac{\Sigma(x - \bar{x})^2}{n_1 - 1} = \frac{636}{7}$$

$$S^2_2 = \frac{\Sigma(y - \bar{y})^2}{n_2 - 1} = \frac{1200}{9} = \frac{400}{3}$$

Since  $S^2_2 > S^2_1$ , the F-statistic is given by

$$F = \frac{S^2_1}{S^2_2} = 1.467$$

Since  $1.467 = F_{\text{cal}} < F_{\text{tab}} = 3.68$  at 5% level of significance for (9,7) degree of freedom. So null hypothesis is accepted.

**Q.5. A random variable X has the following probability function:**

| x    | 0 | 1 | 2  | 3  | 4  | 5              | 6               | 7                   |
|------|---|---|----|----|----|----------------|-----------------|---------------------|
| P(x) | 0 | k | 2k | 2k | 3k | k <sup>2</sup> | 2k <sup>2</sup> | 7k <sup>2</sup> + k |

(i) Find k

(ii) Evaluate  $P(X < 6)$ ,  $P(X \geq 6)$ ,  $P(3 < X \leq 6)$

(iii) Find the minimum value of x so that  $P(X \leq x) > \frac{1}{2}$ .

**Ans. (i)** Since  $\sum_{x=0}^7 p(x) = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$(10k - 1)(k + 1) = 0$$

$$\Rightarrow k = \frac{1}{10}$$

[ $\because p(x) > 0$ ]

(ii)  $P(X < 6) = P(X = 0) + P(X = 1) + \dots + P(X = 5)$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= \frac{8}{10} + \frac{1}{100} = \frac{81}{100}$$

$$P(X \geq 6) = P(X = 6) + P(X = 7)$$

$$= 2k^2 + 7k^2 + k = \frac{9}{100} + \frac{1}{10} = \frac{19}{100}$$

$$P(3 < X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 3k + k^2 + 2k^2$$



$$= \frac{3}{10} + \frac{3}{100} = \frac{33}{100}$$

$$(iii) P(X \leq 1) = k = \frac{1}{10} < \frac{1}{2}$$

$$P(X \leq 2) = k + 2k = \frac{3}{10} < \frac{1}{2}$$

$$P(X \leq 3) = k + 2k + 2k = \frac{5}{10} = \frac{1}{2}$$

$$P(X \leq 4) = k + 2k + 2k + 3k = \frac{8}{10} > \frac{1}{2}$$

$\therefore$  The minimum value of  $x$  so that  $P(X < x) > \frac{1}{2}$  is 4.

**Q.6. X is a continuous random variable with probability density function given by**

$$f(x) = \begin{cases} kx & \text{if } 0 \leq x < 2 \\ 2k & \text{if } 2 \leq x < 4 \\ -kx + 6k & \text{if } 4 \leq x < 6 \end{cases}$$

**Find k and mean value of X.**

**Ans.** Since total probability = 1

$$\text{We have } \int_0^6 f(x) dx = 1$$

$$\int_0^2 f(x) dx + \int_2^4 f(x) dx + \int_4^6 f(x) dx = 1$$

$$\int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (-kx + 6k) dx = 1$$

$$k \left[ \frac{x^2}{2} \right]_0^2 + 2k [x]_2^4 + \left[ -k \frac{x^2}{2} + 6kx \right]_4^6 = 1$$

$$k(2 - 0) + 2k(4 - 2) + (-18k + 36k) - (-8k + 24k) = 1$$

$$\Rightarrow \begin{aligned} 8k &= 1 \\ k &= \frac{1}{8} \end{aligned}$$

$$\text{Mean value of X} = E(X) = \int_0^6 x f(x) dx$$

$$= \int_0^2 x \cdot kx dx + \int_2^4 x \cdot 2k dx + \int_4^6 x(-kx + 6k) dx$$

$$\begin{aligned}
&= k \left[ \frac{x^3}{3} \right]_0^2 + 2k \left[ \frac{x^2}{2} \right]_2^4 + \left[ -k \frac{x^3}{3} + 6k \cdot \frac{x^2}{2} \right]_4^6 \\
&= k \left( \frac{8}{3} \right) + k(12) - \frac{k}{3}(152) + 3k(20) \\
&= 24k = 24 \times \frac{1}{8} = 3
\end{aligned}$$

**Q.7.** The probability of a shooter hitting a target is  $\frac{3}{4}$ . How many minimum number of times must he/she fire so that the probability of hitting the target atleast once is more than 0.99?

**Ans.** Let the shooter fire  $n$  times. Here  $n$  fires are  $n$  Bernoulli trials with

$$P = \frac{3}{4} \text{ \& } q = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{Now } P(X \geq 1) > 0.99 \quad (\text{given})$$

$$1 - P(X = 0) > 0.99$$

$$P(X = 0) < 0.01$$

$${}^nC_0 \left( \frac{1}{4} \right)^n < \frac{1}{100}$$

$$4^n > 100$$

...(1)

The minimum value of  $n$  satisfying. Hence the shooter must fire 4 times.

**Q.8.** Data was collected over a period of 10 years, showing number of deaths from horse kicks in each of the 200 army corps. The distribution of deaths was as follows:

|               |     |    |    |   |   |       |
|---------------|-----|----|----|---|---|-------|
| No. of deaths | 0   | 1  | 2  | 3 | 4 | Total |
| Frequency     | 109 | 65 | 22 | 3 | 1 | 200   |

Fit a poisson distribution to the data and calculate the theoretical frequencies.

**Ans.** Mean of given distribution

$$= \frac{\sum fx}{\sum f} = \frac{65 + 44 + 9 + 4}{200}$$

$$= \frac{122}{200}$$

$$= 0.61$$

This is the parameter ( $m$ ) of the Poisson distribution.

$\therefore$  Required Poisson distribution is  $N \cdot \frac{m^r e^{-m}}{r!}$

$$= 200 e^{-0.61} \cdot \frac{(0.61)^r}{r!} = 108.7 \times \frac{(0.61)^r}{r!}$$



| r | P(r)                                      | Theoretical Frequency |
|---|-------------------------------------------|-----------------------|
| 0 | 108.7                                     | 109                   |
| 1 | $108.7 \times 0.61 = 66.3$                | 66                    |
| 2 | $108.7 \times \frac{(0.61)^2}{2!} = 20.2$ | 20                    |
| 3 | $108.7 \times \frac{(0.61)^3}{3!} = 4.1$  | 4                     |
| 4 | $108.7 \times \frac{(0.61)^4}{4!} = 0.7$  | 1                     |
|   |                                           | Total = 200           |

**Q.9.** The scores of two batsmen A and B in ten innings during a certain season are as follows:

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| A | 32 | 28 | 47 | 63 | 71 | 39 | 10 | 60 | 96 | 14 |
| B | 19 | 31 | 48 | 53 | 67 | 90 | 10 | 62 | 40 | 80 |

Find which of the two batsmen, A & B, is more consistent in scoring.

Ans. Let  
and

$$x_0 = 46$$

$$y_0 = 50$$

| Score of A ( $x_i$ ) | $u = x_i - x_0$<br>$= x_i - 46$ | $u^2_i$ |
|----------------------|---------------------------------|---------|
| 32                   | -14                             | 196     |
| 28                   | -18                             | 324     |
| 47                   | 1                               | 1       |
| 63                   | 17                              | 289     |
| 71                   | 25                              | 625     |
| 39                   | -7                              | 49      |
| 10                   | -36                             | 1296    |
| 60                   | 14                              | 196     |
| 96                   | 50                              | 2500    |
| 14                   | -32                             | 1024    |
| Total                | 0                               | 6500    |

$$S.D. = S_x = S_u = \sqrt{\frac{\sum u_i^2}{N} - \bar{u}}$$

$$= \sqrt{\frac{6500}{10} - \frac{0}{10}} = 25.5$$



|   |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                      |
|---|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 5 | When data is normally distributed, the mean is widely preferred. | When data distribution is skewed, median is the best representative.                                                                                                                                                                                                                                                                                                                   | When there is a nominal distribution of data, the mode is preferred. |
| 6 | Mean<br>$\bar{x} = \Sigma x/N$                                   | <p>If the total number of observations (n) are odd then median is:</p> <p>Median = <math>\left(\frac{n+1}{2}\right)^{th}</math> observation</p> <p>If the total number of observations (n) are even number, then the formula is given below:</p> <p>Median =</p> $\frac{\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{n}{2} + 1\right)^{th} \text{ observation}}{2}$ | The mode is the most frequently occurring observation or value       |

**Q.12. Differentiate between Descriptive and Inferential Statistics.**

**Ans.**

| Basis      | Descriptive Statistics                                                                             | Inferential Statistics                                                                                      |
|------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Definition | Descriptive statistics is used to describe the characteristics of the population using a sample.   | Inferential statistics uses various analytical tools to draw inferences about the population using samples. |
| Tools      | Measures of central tendency and measures of dispersion.                                           | Hypothesis testing and regression analysis.                                                                 |
| Use        | Organizes, describes and presents data in a meaningful way with the help of charts and graphs.     | Tests, predicts, and compares data obtained from various samples.                                           |
| Relevance  | It is used to summarize known data in a way that can be used for further predictions and analysis. | It tries to use the summarized samples to draw conclusions about the population.                            |
| Definition | Descriptive statistics is used to describe the characteristics of the population using a sample.   | Inferential statistics uses various analytical tools to draw inferences about the population using samples. |
| Tools      | Measures of central tendency and measures of dispersion.                                           | Hypothesis testing and regression analysis.                                                                 |

**Q.13. Define the relationship between Dependent and Independent Variables.**

**Ans.** In statistics and research, the relationship between independent and dependent variables is fundamental to understanding the cause-and-effect dynamics within a given phenomenon. Here's a breakdown:



**Independent Variable:** Also known as the predictor variable or explanatory variable, the independent variable is the variable that is manipulated or controlled by the researcher. It is the presumed cause or influence in the relationship. For example, in a study investigating the effect of study time on exam scores, study time would be the independent variable.

**Dependent Variable:** Also known as the response variable, the dependent variable is the variable being measured or observed in an experiment. It is the outcome or result that is presumed to be influenced by changes in the independent variable. Using the same example, exam scores would be the dependent variable.

The relationship between the two can be summarized as follows:

1. Changes in the independent variable are hypothesized or expected to cause changes in the dependent variable.
2. The nature and strength of this relationship can vary. It could be linear, non-linear, positive, negative, or have no discernible pattern.
3. The relationship between the independent and dependent variables is often explored through statistical analysis techniques like regression analysis, correlation analysis, or other modeling techniques.

Understanding this relationship is crucial for researchers as it helps in making predictions, testing hypotheses, and drawing conclusions about the phenomenon under study. Additionally, identifying and understanding the relationship between these variables is essential for designing effective experiments and studies.

**Q.14. What do you understand by Hypothesis Testing. Also explain types of Hypothesis testing and its four main required steps. (10)**

**Ans.** Hypothesis testing is a statistical method used to determine if there is enough evidence in a sample data to draw conclusions about a population. It involves formulating two competing hypotheses, the null hypothesis ( $H_0$ ) and the alternative hypothesis ( $H_1$ ), and then collecting data to assess the evidence.

**There are three types of hypothesis tests:** right-tailed, left-tailed, and two-tailed. Right-tailed tests assess if a parameter is greater, left-tailed if lesser. Two-tailed tests check for non-directional differences, greater or lesser.

Different types of Hypotheses for some known value 'c'

| Two Sided          | Left Sided      | Right Sided     |
|--------------------|-----------------|-----------------|
| $H_0 : \mu = c$    | $H_0 : \mu = c$ | $H_0 : \mu = c$ |
| $H_1 : \mu \neq c$ | $H_1 : \mu < c$ | $H_1 : \mu > c$ |

**Four main required steps of Hypothesis testing**

**Step 1:** State the hypotheses.

**Step 2:** Set the criteria for a decision.

**Step 3:** Compute the test statistic.

**Step 4:** Make a decision/ State a conclusion.



**STEP 1: State the hypotheses.**

A researcher states a null hypothesis about a value in the population ( $H_0$ ) and an alternative hypothesis that contradicts the null hypothesis.

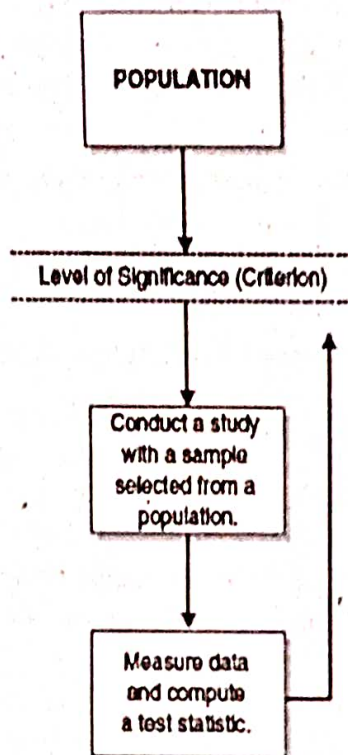
**STEP 2: Set the criteria for a decision.**

A criterion is set upon which a researcher will decide whether to retain or reject the value stated in the null hypothesis.

A sample is selected from the population, and a sample mean is measured.

**STEP 3: Compute the test statistic.**

This will produce a value that can be compared to the criterion that was set before the sample was selected.

**STEP 4: Make a decision.**

If the probability of obtaining a sample mean is less than 5% when the null is true, then reject the null hypothesis.

If the probability of obtaining a sample mean is greater than 5% when the null is true, then retain the null hypothesis.

**Q.15.** Consider the following tally form 11 sample plots on Heiburg Forest, where  $X$ , is the number of downed logs per acre. Compute the basic statistics (Mean, Median, Standard deviation, Range, Coefficient of Variation for the sample plots).

| ID  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |
|-----|----|----|----|----|----|----|----|----|----|----|----|
| $x$ | 25 | 35 | 55 | 15 | 40 | 25 | 55 | 35 | 45 | 5  | 20 |

**Ans.**

| ID | $x_i$ | $x_i^2$ | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$ | Order |
|----|-------|---------|-------------------|---------------------|-------|
| 1  | 25    | 625     | -7.27             | 52.88529            | 4     |
| 2  | 35    | 1225    | 2.73              | 7.4529              | 6     |
| 3  | 55    | 3025    | 22.73             | 516.6529            | 10    |
| 4  | 15    | 225     | -17.25            | 298.2529            | 2     |
| 5  | 40    | 1600    | 7.73              | 59.7529             | 8     |
| 6  | 25    | 625     | -7.27             | 52.88529            | 5     |
| 7  | 55    | 3025    | 22.73             | 516.6529            | 11    |
| 8  | 35    | 1225    | 2.73              | 7.4529              | 7     |
| 9  | 45    | 2025    | 12.73             | 162.0529            | 9     |
| 10 | 5     | 25      | -27.27            | 743.6529            | 1     |
| 11 | 20    | 400     | -12.27            | 150.1819            | 3     |
|    | 355   | 14025   | 0                 | 2568.1519           |       |



1. Sample mean =  $\bar{x} = \sum \frac{x_i}{n} = \frac{355}{11} = 32.27$

2. As number of observation  $n = 11$

Median =  $\left(\frac{n+1}{2}\right)^{th}$  observation =  $\left(\frac{11+1}{2}\right)^{th}$  observation = 6<sup>th</sup> observation = 35

3. Variance =  $s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1} = \frac{2563.1519}{11-1} = 256.32$

4. Standard Deviation =  $s = \sqrt{256.32} = 16.0256$

5. Range = Highest Observation - Lowest Observation =  $55 - 5 = 50$

6. Coefficient of variation =  $CV = \frac{s}{\bar{x}} \times 100 = \frac{16.0256}{32.27} \times 100 = 49.66$

7. Standard error of the mean =  $S_{\bar{x}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{256.32}{11}} = 4.8319$

## UNIT - 2

There are three types of linear models

- (1) Simple linear regression model
- (2) Multiple linear regression model
- (3) Analysis of variance model (ANOVA)

Simple linear regression model can be stated as

$$y = \beta_0 + \beta_1 x + \epsilon$$

Multiple linear regression model can be stated as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \epsilon$$

Sum of squared error

$$SSE = \sum (y_i - \hat{y}_i)^2$$

$$SSE = \begin{cases} \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 & \text{for simple linear regression model} \\ \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_k x_{ki})^2 & \text{for multiple linear regression model} \end{cases}$$

Properties of the least squares estimators

- (a) The least squares estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are unbiased i.e.,

$$E[\hat{\beta}_1] = \beta_1$$

and

$$E[\hat{\beta}_0] = \beta_0$$

- (b) Variance of regression parameters

$$\text{Var}[\hat{\beta}_1] = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

$$\text{Var}[\hat{\beta}_0] = \sigma^2 \left[ \frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right]$$

Regression sum of squares (SSR)

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

Total sum of squares (SST)

$$SST = SSR + SSE = \sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{SSE}{SST}$$



**ANOVA Models**• **Single Factor Anova Model**

$$y_1 = \mu + t_1 + \epsilon_1$$

$$y_2 = \mu + t_2 + \epsilon_2$$

• **Sum of squares total (SST)**

$$SST = \sum \sum (y_{ij} - \bar{y})^2 = \sum \sum (y_{ij}^2) - \left(\frac{G^2}{n}\right)$$

where

$$G = \sum \sum y_{ij}$$

• **Sum of squares between treatments (SS(tr))**

$$SS(tr) = \sum \left( \frac{y_i^2}{n_i} \right) - \frac{G^2}{n}$$

• **Sum of square total (SST)**

$$SST = SS(tr) + SSE = \sum \sum (y_{ij} - \bar{y}_i)^2$$

• **Mean sum of square between treatments (MSS(tr))**

$$MSS(tr) = \frac{SS(tr)}{df} = \frac{SS(tr)}{r-1}$$

• **Mean sum of square within treatment (MSE)**

$$MSE = \frac{SSE}{df} = \frac{SSE}{n-r}$$

• **F - ratio**

$$F = \frac{\text{larger variance}}{\text{smaller variance}} = \frac{MSS(tr)}{MSE}$$

Two factor ANOVA model can be expressed as

$$y_{ij} = \mu + \alpha_i + \beta_j + \mu_{ij}; \quad i = 1, 2 \text{ \& } j = 1, 2$$

where  $y_{ij}$  is the weight gain of the  $(i, j)^{\text{th}}$  value and  $\epsilon_{ij}$  is a random error.• **Sum of squares total (SST)**

$$SST = \sum y_{ij}^2 - \frac{G^2}{n} \text{ where } G = \sum \sum y_{ij}$$

• **Sum of squares to column (SSC)**

$$SSC = \frac{(\sum y_1)^2}{n_1} + \frac{(\sum y_2)^2}{n_2} + \dots - \frac{G^2}{n}$$

where  $\sum y_1, \sum y_2, \dots$  represent column total

- Sum of squares of rows (SSR)

$$SSR = \frac{(\Sigma y_1)^2}{n_1} + \frac{(\Sigma y_2)^2}{n_2} + \dots - \frac{G^2}{n}$$

where  $\Sigma y_1, \Sigma y_2, \dots$  represent row total.

- Sum of squares errors (SSE)

$$SSE = SST - (SSC + SSR)$$

- Mean sum of squares of columns (MSC)

$$MSC = \frac{SSC}{df} = \frac{SSC}{C-1}$$

- Mean sum of squares of rows (MSR)

$$MSR = \frac{SSR}{df} = \frac{SSR}{r-1}$$

$$MSE = \frac{SSE}{df} = \frac{SSE}{(r-1)(c-1)}$$

- F - ratio in respect of column

$$F_C = \frac{MSC}{MSE}$$

**Q.1.** Three experimenters determine the moisture content of samples of body lotion. For this purpose, each experimenter has taken 4 consignments. The results are given below:

| Experimenter | Consignments |    |     |    |
|--------------|--------------|----|-----|----|
|              | I            | II | III | IV |
| A            | 9            | 10 | 9   | 10 |
| B            | 12           | 11 | 9   | 11 |
| C            | 11           | 12 | 10  | 12 |

Test whether there is any significant difference among consignments and among experimenters at 5% significance of level.

Ans.

| Experimenter | Consignments |    |     |    |                |
|--------------|--------------|----|-----|----|----------------|
|              | I            | II | III | IV | Sum of columns |
| A            | 9            | 10 | 9   | 10 | 38             |
| B            | 12           | 11 | 9   | 11 | 43             |
| C            | 11           | 12 | 10  | 12 | 45             |
| Sum of rows  | 32           | 33 | 28  | 33 | 126            |

$H_0$ : There is no significant difference between consignments.

There is no significant difference between experimenters.

$H_1$ : There is significant difference between consignments

There is significant difference between experimenters.



$$\begin{aligned}
 SST &= \sum y_{ij}^2 - \frac{G^2}{n} \\
 &= 9^2 + 10^2 + 9^2 + 10^2 + 12^2 + 11^2 + 9^2 + 11^2 \\
 &\quad + 11^2 + 12^2 + 10^2 + 12^2 - \frac{(126)^2}{12} = 15.0
 \end{aligned}$$

$$SSC = \frac{(\sum y_1)^2}{n_1} + \frac{(\sum y_2)^2}{n_2} + \dots - \frac{G^2}{n}$$

$$\frac{32^2}{4} + \frac{33^2}{4} + \frac{28^2}{4} + \frac{33^2}{4} - \frac{126^2}{12} = 5.67$$

$$\begin{aligned}
 SSR &= \frac{(\sum y_1)^2}{n_1} + \frac{(\sum y_2)^2}{n_2} + \dots - \frac{G^2}{n} \\
 &= \frac{38^2}{4} + \frac{43^2}{4} + \frac{45^2}{4} - \frac{126^2}{12} = 6.5
 \end{aligned}$$

$$\begin{aligned}
 SSE &= SST - (SSC + SSR) \\
 &= 15 - 5.67 - 6.5 = 2.83
 \end{aligned}$$

$$MSC = \frac{SSC}{df} = \frac{SSC}{C-1} = \frac{5.67}{3} = 1.89$$

$$MSR = \frac{SSR}{df} = \frac{6.5}{2} = 3.25$$

$$\begin{aligned}
 MSE &= \frac{SSE}{df} = \frac{SSE}{(r-1)(c-1)} = \frac{2.83}{3 \times 2} \\
 &= 0.47
 \end{aligned}$$

$$F_C = \frac{1.89}{0.47} = 4.0$$

$$F_r = \frac{3.25}{0.47} = 6.89$$

If  $\alpha = 5\%$ ,  $F_{0.5}(2, 15) = 5.14$

If  $\alpha = 5\%$ ,  $F_{0.5}(3, 15) = 4.76$

ANOVA Table

| Source of Variation   | Sum of Squares | Degree of freedom | Mean square | F Value | $F_{\alpha}(r-1, n-r)$ |
|-----------------------|----------------|-------------------|-------------|---------|------------------------|
| Between Experimenters | 6.5            | 2                 | 3.25        | 6.89    | 5.14                   |
| Within Consignments   | 5.67           | 3                 | 1.89        | 4.0     | 4.76                   |
| Error                 | 2.83           | 6                 | 0.47        |         |                        |
| Total                 | 15             | 11                |             |         |                        |



The F ratio 6.89 ( $> 5.14$ ) does lie in the critical region at 5% level of significance, therefore, we conclude that the mean moisture content as determined by 3 experimenters are not equal. The F-ratio  $4.0 < 4.76$  does lie in the critical region at 5% level of significance therefore, we conclude that the moisture content of 4 consignment may not differ from each other.

**Q.2. Write a short note on Logistic regression.**

**Ans.** Logistic regression is a statistical technique for describing and explaining the connection between one dependent binary variable and one or more nominal, ordinal, interval, or ratio-level independent variables.

Types of logistic regression

- Binary logistic regression
- Multinomial logistic regression
- Ordinal logistic regression

**Binary Logistic Regression:**

Based on the values of the independent variables, binary logistic regression is used to estimate the likelihood of being a case (predictors).

**Multinomial logistic regression**

It is a classification technique that extends logistic regression to situations with more than two discrete outcomes.

**Ordinal Logistic Regression**

It is an extension of binary logistic regression.

With ordered multiple categories and independent variables, ordinal regression is used to predict the dependent variables.

**Q.3. Discuss the geometry of least square method.**

**Ans.** The least squares method is a mathematical technique commonly used in data fitting, linear regression and optimization problems. It aims to minimize the sum of the squared differences (residuals) between observed values and estimated values. From a geometric perspective the least squares method has an intuitive interpretation especially when applied to linear regression.

The geometry of the least squares method involves projecting observed outcomes onto the column space of the design matrix, with the resulting projection minimizing the squared differences (residuals) between observed and estimated outcomes. The orthogonality between the residual vector and the column space is key to understanding why the projection minimizes the sum of squared residuals. This geometric interpretation provides insight into the mechanics and optimality of the least squares approach.

**Q.4. What do you mean by multiple linear regression? Also find the multiple linear regression model for the following data.**

| y   | $x_1$ | $x_2$ |
|-----|-------|-------|
| 140 | 60    | 22    |
| 155 | 62    | 25    |
| 159 | 67    | 24    |
| 179 | 70    | 20    |



|     |    |    |
|-----|----|----|
| 102 | 71 | 15 |
| 200 | 72 | 14 |
| 212 | 75 | 14 |
| 215 | 78 | 11 |

**Ans.** Multiple linear regression is a method we can use to quantify the relationship between two or more predictor variables and a response variable.  
For multiple linear regression model

| y    | X <sub>1</sub> | X <sub>2</sub> | X <sub>1</sub> <sup>2</sup> | X <sub>2</sub> <sup>2</sup> | X <sub>1</sub> y | X <sub>2</sub> y | X <sub>1</sub> X <sub>2</sub> |
|------|----------------|----------------|-----------------------------|-----------------------------|------------------|------------------|-------------------------------|
| 140  | 60             | 22             | 3600                        | 484                         | 8400             | 3080             | 1320                          |
| 155  | 62             | 25             | 3844                        | 625                         | 9610             | 3875             | 1550                          |
| 159  | 67             | 24             | 4489                        | 576                         | 10653            | 3816             | 1608                          |
| 179  | 70             | 20             | 4900                        | 400                         | 12530            | 3580             | 1400                          |
| 192  | 71             | 15             | 5041                        | 225                         | 13632            | 2880             | 1065                          |
| 200  | 72             | 14             | 5184                        | 196                         | 14400            | 2800             | 1008                          |
| 212  | 75             | 14             | 5625                        | 196                         | 15900            | 2968             | 1050                          |
| 215  | 78             | 11             | 6084                        | 121                         | 16770            | 2365             | 858                           |
| 1452 | 555            | 145            | 38767                       | 2823                        | 101895           | 25364            | 9859                          |

$$\bar{y} = 181.5$$

$$\bar{X}_1 = 69.375$$

$$\bar{X}_2 = 18.125$$

Calculate regression sums

$$\Sigma x_1^2 = \Sigma X_1^2 - \frac{(\Sigma X_1)^2}{n} = 38767 - \frac{(555)^2}{8} = 263.875$$

$$\Sigma x_2^2 = \Sigma X_2^2 - \frac{(\Sigma X_2)^2}{n} = 2823 - \frac{(145)^2}{8} = 194.875$$

$$\begin{aligned} \Sigma x_1 y &= \Sigma X_1 y - \frac{(\Sigma X_1 \Sigma y)}{n} \\ &= 101895 - \frac{(555 \times 1452)}{8} = 1162.5 \end{aligned}$$

$$\begin{aligned} \Sigma x_2 y &= \Sigma X_2 y - \frac{(\Sigma X_2 \Sigma y)}{n} \\ &= 25364 - \frac{(145 \times 1452)}{8} = -953.5 \end{aligned}$$

$$\begin{aligned} \Sigma x_1 x_2 &= \Sigma X_1 X_2 - \frac{(\Sigma X_1 \Sigma X_2)}{n} \\ &= 9859 - \frac{(555 \times 145)}{8} = -200.375 \end{aligned}$$

Calculate  $b_0$ ,  $b_1$  and  $b_2$

$$b_1 = \frac{[(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)]}{[(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2]}$$

$$= \frac{[(194.875)(1162.5) - (-200.375)(-953.5)]}{[(263.875)(194.875) - (-200.375)^2]}$$

$$= 3.148$$

$$b_2 = \frac{[(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)]}{[(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2]}$$

$$b_2 = \frac{[(263.875)(-953.5) - (-200.375)(1152.5)]}{[(263.875)(194.875) - (-200.375)^2]}$$

$$= -1.656$$

$$b_0 = \bar{y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

$$= 181.5 - (3.148)(69.375) - (-1.656)(18.125)$$

$$= -6.867$$

Estimated linear regression equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2$$

$$\hat{y} = 6.867 + 3.148 x_1 - 1.656 x_2$$

#### Q.5. Discuss poisson regression model.

**Ans.** Poisson regression is a statistical technique used to model count data or data representing the number of events occurring within a fixed interval or space. It is especially useful when the response variable represents counts that are discrete, non-negative and often occur infrequently or over varying time periods.

Poisson regression is based on the poisson distribution. In a poisson regression model, the expected count (mean) is modeled as a function of one or more predictor variables. The relationship is typically expressed in terms of the logarithm of the expected count, making poisson regression a type of generalized linear model (GLM).

#### Mathematical Formulation

Given a set of predictor variables  $x_1, x_2, \dots, x_k$ , the general form of the poisson regression model is:

$$\log(\lambda) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

where

- $\lambda$  is the expected count
- $\beta_0, \beta_1, \dots, \beta_k$  are the coefficients corresponding to the predictor variables.
- $x_1, x_2, \dots, x_k$  are the predictor variables
- The logarithmic link function ensures that the expected count remains positive.

#### Q.6. Define ANCOVA model and explain one-way model with one covariate.

**Ans.** The analysis of covariance (ANCOVA) model

In general, an analysis of covariance model can be written as

$$y = \bar{Z}\alpha + X\beta + \epsilon$$



where  $Z$  contains  $0$ s and  $1$ s &  $\alpha$  contains  $\mu$  and parameters such as  $\alpha_i, \beta_j$  and  $(\alpha\beta)_{ij}$  representing factor and interactions (or other effects);  $X$  contains the covariate values; and  $\beta$  contains coefficients of the covariates. We use  $X\beta$  to represent the covariates in the model.

One-way model with one covariate

The one-way (balanced) model can be expressed as;

$$y_{ij} = \mu + \alpha_i + \beta x_{ij} + \epsilon_{ij}; i = 1, 2, \dots, k; j = 1, 2, \dots, n$$

All  $kn$  observations can be written in the form of

$$y = Za + X\beta + C$$

The parameters  $\alpha$  can be estimated as;

$$\hat{a} = \hat{a}_0 - (Z^T Z)^{-1} Z^T X \hat{\beta}$$

$$\hat{a}_0 = (Z^T Z)^{-1} Z^T y \text{ is a solution}$$

For the normal equations for the model

$$y = Za + \epsilon \text{ without the covariates.}$$

The estimator of  $\beta$  is:

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

where

$$S_{xx} = \sum_{i=1}^k \sum_{j=1}^n (x_{ij} - \bar{x}_i)^2$$

$$S_{xy} = \sum_{i=1}^k \sum_{j=1}^n (x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i)$$

$$S_{yy} = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_i)^2$$

For the analysis of covariance model we denote SSE as  $SSE_{y,x}$

$$SSE_{y,x} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

$$= \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_i)^2 - \frac{\left( \sum_{i=1}^k \sum_{j=1}^n (x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i) \right)^2}{\sum_{i=1}^k \sum_{j=1}^n (x_{ij} - \bar{x}_i)^2}$$

which has  $k(n-1)-1$  degree of freedom

The degree of freedom of  $SSE_{y,x}$  are reduced by 1 for estimation of  $b$ , since

$SSE_y = SS_{yy}$  has  $k(n-1)$  degree of freedom and  $\frac{S_{xy}^2}{S_{xx}}$  has 1 degree of freedom.

In using analysis of covariance the researcher expects the reduction from  $SSE_y$  to  $SSE_{y,x}$  to at least offset the loss of a degree of freedom.

**Q.7.** There are two series of index numbers, P for price index and S for stock of commodity. The mean and standard deviation of P are 100 and 8 and S are 103 and 4 respectively. The correlation coefficient between the two series is 0.4. With these data, work out a linear equation to read off values of P for various values of S. Can the same equation be used to read off values of S for various values of P?

**Ans.** Let us assume that P = price index be variable X and S = stock of commodity be variable Y.

Linear equation to read off values of P for various values of S would be regression equation of X on Y. Regression co-efficient is to be computed using mean and standard deviation.

Given that

$$\bar{X} = 100, \bar{Y} = 103$$

$$\sigma_x = 8, \sigma_y = 4, r = 0.4$$

Regression equation of X on Y:

$$(X - \bar{X}) = b_{xy} (Y - \bar{Y})$$

$$(X - \bar{X}) = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y})$$

$$(X - 100) = \left(0.4 \times \frac{8}{4}\right) (Y - 103)$$

$$X = 17.6 + 0.8 Y$$

Linear equation to read off values of P for various values of S is  $X = 17.6 + 0.8 Y$

To read off values of S for various values of P we need regression equation of Y on X and therefore above linear equation cannot be used. Hence the following regression equation of Y on X be computed.

$$(Y - \bar{Y}) = b_{yx} (X - \bar{X})$$

$$(Y - \bar{Y}) = r \frac{\sigma_y}{\sigma_x} (X - \bar{X})$$

$$(Y - 103) = \left(0.4 \times \frac{4}{8}\right) (X - 100)$$

$$Y = 83 + 0.2 X$$

Hence the linear equation to read off values of S for various values of P is

$$Y = 83 + 0.2 X$$

**Q.8.** From the following data obtain the two regression lines:

|   |   |   |   |   |    |   |    |    |
|---|---|---|---|---|----|---|----|----|
| X | 7 | 8 | 5 | 9 | 12 | 9 | 10 | 15 |
| Y | 4 | 5 | 2 | 6 | 9  | 5 | 7  | 12 |

**Ans.**

| X | Y | X <sup>2</sup> | Y <sup>2</sup> | XY |
|---|---|----------------|----------------|----|
| 7 | 4 | 49             | 16             | 28 |



|    |    |     |     |     |
|----|----|-----|-----|-----|
| 8  | 5  | 64  | 25  | 40  |
| 5  | 2  | 25  | 4   | 10  |
| 9  | 6  | 81  | 36  | 54  |
| 12 | 9  | 144 | 81  | 108 |
| 9  | 5  | 81  | 25  | 45  |
| 10 | 7  | 100 | 49  | 70  |
| 15 | 12 | 225 | 144 | 180 |
| 75 | 50 | 769 | 324 | 525 |

$$\bar{X} = \frac{\sum X}{n} = \frac{75}{8} = 9.375$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{50}{8} = 6.25$$

$$b_{xy} = \frac{n \sum XY - \sum X \sum Y}{n \sum Y^2 - (\sum Y)^2} = \frac{8(525) - (75)(50)}{8(324) - (50)^2}$$

$$= 0.9815$$

$$b_{yx} = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$= \frac{8(525) - (75)(50)}{8(769) - (75)^2}$$

$$= 1.0057$$

Regression line X on Y

$$(X - \bar{X}) = b_{xy} (Y - \bar{Y})$$

$$X - 9.375 = 0.9815 (Y - 6.25)$$

$$\boxed{X = 3.2406 + 0.9815Y}$$

Regression line Y on X

$$(Y - \bar{Y}) = b_{yx} (X - \bar{X})$$

$$Y - 6.25 = 1.0057 (X - 9.375)$$

$$Y = -3.1784 + 1.0057 X$$

**Q.9. Define confounding variable.**

**Ans.** Confounding variables, also known as confounders, are additional variables that are not the main independent or dependent variables of interest in a study but still have an effect on the dependent variable. They can create confusion or distortion in the interpretation of the relationship between the independent and dependent variables. Confounding variables are associated with both the independent and dependent variables. They can obscure the true relationship between the independent variable and the dependent variable because they introduce additional variability.



**Q.10. Proof Gauss Markov Theorem. Elaborate it with proper proofing.**

**Ans.** Gauss-Markov Theorem states that

If  $E[y] = X\beta$  and  $\text{cov}(y) = \sigma^2 I$ , the least-squares estimators  $\hat{\beta}_j$  for all  $j = 0, 1, 2, 3, \dots, k$  have minimum variance among all linear unbiased estimators.

**Proof:** We examine a linear estimator  $Ay$  for  $\beta$  and aim to determine the matrix  $A$  that result in  $Ay$  being a minimum variance unbiased estimator of  $\beta$ . For  $Ay$  to qualify as an unbiased estimator of  $\beta$ . It's necessary that:

$$E[Ay] = \beta$$

Using the assumption  $E[y] = \beta$ , this can be expressed as:

$$E[Ay] = AE[y] = AX\beta = \beta,$$

Which gives the unbiased condition  $AX = I$

Since the relationship  $AX\beta = \beta$  must hold for any possible value of  $\beta$ .

The covariance matrix for the estimator  $Ay$  is given by

$$\text{cov}(Ay) = A(\sigma^2 I)A^T = \sigma^2 AA^T$$

The variances of the  $\hat{\beta}_j$ 's are specifically located along the diagonal of the  $\sigma^2 AA^T$  matrix. Consequently, our goal is to select matrix  $A$ , while adhering to the constraint  $AX = I$ , in such a way that minimizes the diagonal elements of  $AA^T$ .

To relate  $Ay$  and  $\hat{\beta} = (X^T X)^{-1} X^T Y$ , we apply an operation of addition and subtraction involving  $(X^T X)^{-1} X^T$ , resulting in the following expression:

$$AA^T = (A - (X^T X)^{-1} X^T + (X^T X)^{-1} X^T) (A - (X^T X)^{-1} X^T + (X^T X)^{-1} X^T)^T$$

Expanding this expression with respect to  $A - (X^T X)^{-1} X^T$  and  $(X^T X)^{-1} X^T$ , we obtain four terms. Notably, two of these terms become zero due to the constraint  $AX = I$ . Consequently, the result is :

$$AA^T = (A - (X^T X)^{-1} X^T) (A - (X^T X)^{-1} X^T)^T + (X^T X)^{-1}$$

The matrix  $(A - (X^T X)^{-1} X^T) (A - (X^T X)^{-1} X^T)^T$  is positive semidefinite and the diagonal elements are greater than equal to zero. These diagonal elements can be made equal to zero by choosing  $A = (X^T X)^{-1} X^T$ . This value of  $A$  also satisfies the unbiasedness condition  $AX = I$ . The resulting minimum variance estimator of  $\beta$  is  $Ay = (X^T X)^{-1} X^T y$  which is equal to the least-squares estimator of  $\hat{\beta}$ .

**Q.11. What is the fundamental purpose of Simple Linear regression.**

**Ans.** The fundamental purpose of simple linear regression is to establish and quantify the relationship between two variables: an independent variable (predictor variable) and a dependent variable (outcome variable). In simple linear regression, we aim to find a linear equation that best fits the relationship between these variables.

The equation for simple linear regression is typically represented as:

$$\text{Where:} \quad Y = \beta_0 + \beta_1 X + \epsilon$$

- $Y$  is the dependent variable.
- $X$  is the independent variable.
- $\beta_0$  is the intercept (the value of  $Y$  when  $X$  is zero).
- $\beta_1$  is the slope of the line (the change in  $Y$  for a one-unit change in  $X$ ).
- $\epsilon$  represents the error term, which captures the difference between the observed and predicted values of  $Y$ .



The goal of simple linear regression is to estimate the values of  $\beta_0$  and  $\beta_1$  that minimize the sum of squared differences between the observed values of  $Y$  and the values predicted by the linear equation. Once we have estimated the parameters, we can use the regression equation to make predictions about the dependent variable based on values of the independent variable.

In summary, the fundamental purpose of simple linear regression is to model and understand the relationship between two variables, enabling prediction and inference based on this relationship.

**Q.12. What are the two main parameters estimated in Simple Linear Regression.**

**Ans.** Estimation of the regression parameters ( $\beta_0, \beta_1$ ) can be obtained by the method of least squares. In the method of least squares the point estimates of  $\beta_0$  and  $\beta_1$  are the values that minimize the sum of squared error (SSE) between observed values  $y$  and predicted values  $\hat{y}$ :

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$\Sigma$  represents the sum over all data points ( $i = 1$  to  $n$ )

$y_i$  is the observed value for the  $i^{\text{th}}$  data point.

$\hat{y}_i$  is the predicted value for the  $i^{\text{th}}$  data point.

To find the minimum of SSE, we take partial derivatives with respect to  $\beta_0$  and  $\beta_1$  and set them equal to zero.

$$\frac{\partial SSE}{\partial \beta_0} = -2 \sum (y_i - \hat{y}_i) = 0$$

$$\frac{\partial SSE}{\partial \beta_1} = -2 \sum x_i (y_i - \hat{y}_i) = 0$$

Solving for  $\hat{\beta}_0$  and  $\hat{\beta}_1$

Solving the above equations gives us the least squares estimates for  $\beta_0$  and  $\beta_1$ .

For  $\hat{\beta}_0$ :

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Where:

$\bar{y}$  is the mean of the observed values ( $y$ ).

$\bar{x}$  is the mean of the observed values ( $x$ ).

For  $\hat{\beta}_1$ :

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2}$$



These equations provide the estimated coefficients  $\hat{\beta}_0$  and  $\hat{\beta}_1$  that define the best fitting linear regression line, minimizing the sum of squared differences between the observed and predicted values. These estimators are used to make predictions and describe the linear relationship between the predictor variable (x) and the response variable (y).

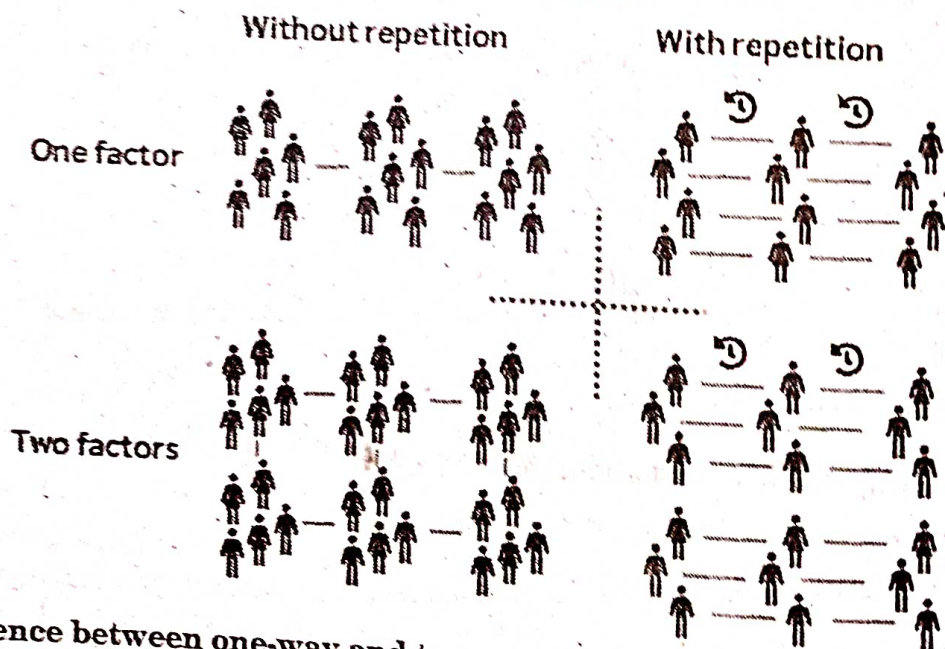
**Q.13. Explain the Concept of ANOVA in detail.**

**Ans.** An analysis of variance (ANOVA) tests whether statistically significant differences exist between more than two samples. For this purpose, the means and variances of the respective groups are compared with each other. In contrast to the t-test, which tests whether there is a difference between two samples, the ANOVA tests whether there is a difference between more than two groups.

There are different types of analysis of variance, being the one-way and two-way analyses of variance the most common ones, each of which can be calculated either with or without repeated measurements.

There are four types of analysis of variance

- One-factor (or one-way) ANOVA
- Two-factors (or two-way) ANOVA
- One-factor ANOVA with repeated measurements
- Two-factors ANOVA with repeated measurements

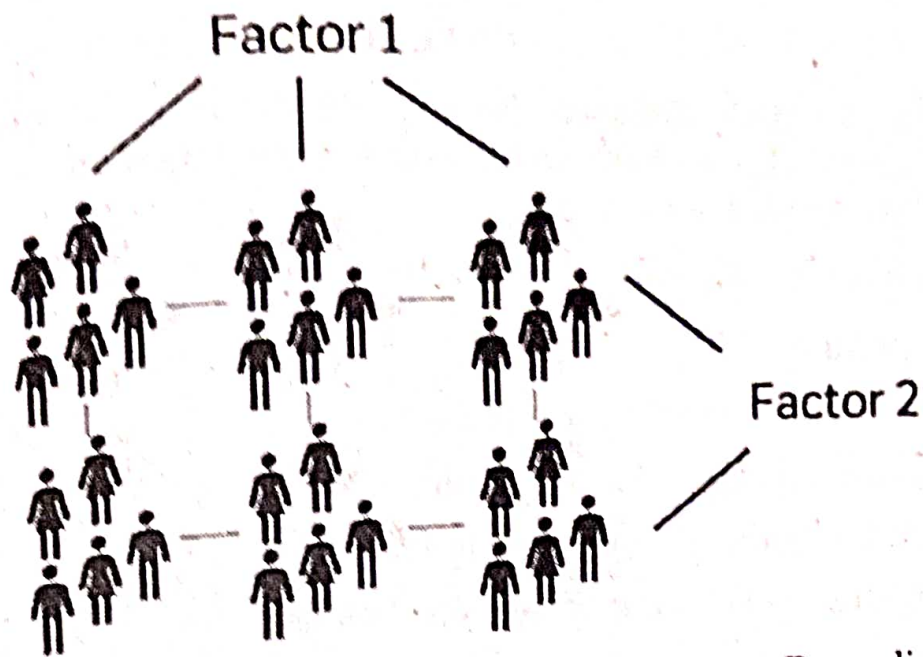


**Difference between one-way and two-way ANOVA**

The one-way analysis of variance only checks whether an independent variable has an influence on a metric dependent variable. This is the case, for example, if it is to be examined whether the place of residence (independent variable) has an influence on the salary (dependent variable). However, if two factors, i.e. two independent variables, are considered, a two-way analysis of variance must be used.

Two-way analysis of variance tests whether there is a difference between more than two independent samples that are split between two variables or factors.





Analysis of variance with and without repeated measures Depending on whether the sample is independent or dependent, either analysis of variance with or without repeated measures is used. If the same person was interviewed at several points in time, the sample is a dependent sample and analysis of variance with repeated measurements is used.

## UNIT - 3

**Open set:** A set  $U \subset X$  is called open, if for each  $x \in U$ ,  $\exists$  an  $\epsilon > 0$  such that the interval  $(x - \epsilon, x + \epsilon)$  is contained in  $U$ . Such an interval is often called an  $\epsilon$ -neighbourhood of  $x$  or simply a neighbourhood of  $x$ .

**Closed set:** A set  $F \subset X$  is called closed if the complements of  $F$ ,  $X \setminus F$ , is open.

**Important points:**

1. Every union of open sets is again open.
2. Every intersection of closed sets is again closed.
3. Every finite intersection of open sets is again open.
4. Every finite union of closed sets is again closed.

**Compact set:** In math, compactness is a property that tries to generalize the idea of a closed & bounded subset in Euclidean space.

A closed and bounded set is known as compact set.

**In general topology**

A subset  $K$  of a topological space is said to be compact if every open cover of  $K$  has a finite subcover.

**Key properties of compact sets**

1. In Euclidean spaces a subset  $K$  is compact iff it is closed and bounded. This is known as the Heine-Borel Theorem.
2. The essential property of compactness is the existence of a finite sub-cover for every open cover.
3. In Euclidean spaces, compactness implies closed and bounded property. However, in more general topological spaces, compact sets might not exhibit these properties.
4. If  $f: X \rightarrow Y$  is a continuous map between topological spaces and  $X$  is compact, then the image  $f(X)$  is compact in  $Y$ .

**Metric (Euclidean distance):** The metric between two points  $x, y \in \mathbb{R}^n$  is defined as

$$\sqrt{\sum_{i=1}^n |x_i - y_i|^2}$$

**Norm:** A norm on  $V$  (vector space) is a function  $|| \cdot ||: V \rightarrow \mathbb{R}$  that satisfies the following properties for all vectors  $x, y \in V$  and all scalars  $\alpha \in \mathbb{R}$ :

1. Non negativity

$$||x|| \geq 0 \text{ and } ||x|| = 0 \text{ iff } x = 0$$

2. Scalar multiplication

$$||\alpha x|| = |\alpha| \cdot ||x||$$

3. Triangle inequality

$$||x + y|| \leq ||x|| + ||y||$$

**Metric space:** A metric space is a set  $X$  equipped with a function  $d: X \times X \rightarrow \mathbb{R}$  called a metric, which satisfies the following properties for all points  $x, y, z$  in  $X$ :



1. Non negativity:  $d(x, y) \geq 0$

2. Identity of indiscernibles:

$$d(x, y) = 0 \text{ iff } x = y$$

3. Symmetry:  $d(x, y) = d(y, x)$

4. Triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z)$$

### Cauchy Sequence

Let  $\{a_j\}_{j=1}^{\infty}$  be a sequence of real (complex) numbers. We say that the sequence satisfies the Cauchy criterion if for each  $\epsilon > 0$  there is an integer  $N > 0$  such that if  $j, k > N$  then  $|a_j - a_k| < \epsilon$

### Completeness in Metric spaces

A metric space is complete if every Cauchy sequence has a limit. In other words, we can say that for every Cauchy sequences  $x_n$  in the metric space, there exists a point  $x$  in the spaces such that  $x_n$  converges to  $x$  as  $n \rightarrow \infty$ .

**Connected metric space:** Let  $(X, d)$  be a metric space:  $X$  is said to be connected if there do not exist two non empty & disjoint open sets  $A$  and  $B$  in  $X$  such that  $X = A \cup B$ . If there exists two disjoint non-empty open sets  $A$  and  $B$  such that  $X = A \cup B$ , then  $X$  is called disconnected and the pair  $\{A, B\}$  is called a disconnection of  $X$ .

**Q.1. Prove that a metric space  $X$  is connected iff every continuous function  $f$  on  $X$  to the discrete metric space  $[1, -1]$  is a constant function.**

**Ans.** Let  $Y$  be the discrete metric space  $\{1, -1\}$ . Let  $X$  be connected and  $f : X \rightarrow \{+1, -1\}$  be continuous. Suppose that  $f$  is not a constant. Let  $A = f^{-1}\{1\}$  and  $B = f^{-1}\{-1\}$ . Then  $A$  and  $B$  are disjoint non-empty open subsets of  $X$  satisfying  $X = A \cup B$ . This is a contradiction to the fact that  $X$  is connected. Therefore  $f$  is constant.

**To get the reverse implication,** let us assume that  $X$  is not connected. Let  $\{A, B\}$  be a disconnection of  $X$ . Now define a map  $f : X \rightarrow Y$  as

$$f(x) = \begin{cases} 1 & \text{if } x \in A \\ -1 & \text{if } x \in B \end{cases}$$

Then  $f$  is continuous because the only open subsets of  $Y$  are  $\emptyset$ ,  $\{1\}$ ,  $\{-1\}$  and  $Y$ . Also  $f$  is not a constant.

**Q.2. Consider the sequence  $\{a_j\}_{j=1}^{\infty}$ . It converges to zero prove it.**

$$\text{Ans. } \{a_j\}_{j=1}^{\infty} = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$$

which seems to indicate that the terms are getting closer and closer to zero. According to the definition of convergence, we need to show that no matter which  $\epsilon > 0$  one chooses, the sequence will eventually become smaller than this number. To

be precise take any  $\epsilon > 0$ . Then there exists a positive integer  $N$  such that  $\frac{1}{N} < \epsilon$ . Therefore, for any  $j > N$  we have.



$$\left| \frac{1}{j} - 0 \right| = \left| \frac{1}{j} \right| < \frac{1}{N} < \epsilon$$

Whenever  $j > N$ .

**Q.3. Prove that  $\mathbb{R}$  and  $\emptyset$  are open and closed simultaneously.**

**Ans.** Take any point  $x$  contained in  $\mathbb{R}$ . Clearly, any neighbourhood of  $x$  consists of real numbers, and hence is also contained in  $\mathbb{R}$ . Therefore  $\mathbb{R}$  is open.

- Take the empty set. Since it does not contain any point, we do not have to check whether any neighbourhoods are contained in the empty set. It is therefore open vacuously.

- Since the complement of the real line is the empty set, and the empty set is open the real line is also closed.

- Since the complement of the empty set is the real line, and the real line is open the empty set is also closed.

**Q.4. Which of the intervals  $(-4, 4)$ ,  $[3, 8]$ ,  $(-4, 5]$ ,  $(0, \infty)$  and  $[0, \infty)$  are open closed, both or neither?**

**Ans.** The interval  $(-4, 4)$  is open because if  $x$  is any number in  $(-4, 4)$  then  $-4 < x < 4$  or equivalently  $-4 - x < 0 < 4 - x$ . Now let  $\epsilon = \min(4 + x, 4 - x)$ . Then  $\epsilon > 0$ , and the interval  $\left(x - \frac{\epsilon}{2}, x + \frac{\epsilon}{2}\right)$  is contained in  $(-4, 4)$ .

- The interval  $[3, 8]$  is closed because its complement consists of the two open sets  $(-\infty, 3)$  &  $(8, \infty)$ .

- The interval  $(-4, 5]$  is neither open nor closed. It is not open because the point  $x = 5$  is contained in the set, but every neighbourhood of that point is not contained inside the set. It is not closed, because its complement  $(-\infty, -4]$  and  $(5, \infty)$  is not open at  $x = -4$ .

- The interval  $(0, \infty)$  and also the interval  $(-\infty, 0)$  are both open, because every point contains a neighbourhood contained inside the set.

- The interval  $[0, \infty)$  and also the interval  $(-\infty, 0]$  are both closed, because their complements are the open sets mentioned above.

**Q.5. Prove that every compact metric space is also complete metric space.**

**Ans.** Suppose  $(K, d)$  is compact. Let  $\{x_n\}_{n=1}^{\infty}$  be a Cauchy sequence in  $K$ . We show that  $(x_n)$  converges thus proving  $K$  is complete. This sequence has a subsequence that converges to some element  $z$  in  $K$ . The sequence itself must converge to  $z$ .

Thus, we can get a full characterization of compactness by adding an additional property to completeness. It can be stated that every compact metric space is "totally bounded". In fact, a metric space is compact iff it is both complete and totally bounded.



**Q.6. Write some examples of connected metric spaces**

**Ans.**

(1) **Closed interval in  $\mathbb{R}$ :** The closed interval  $[a, b]$  in the real line  $\mathbb{R}$  is a connected metric space. Any attempt to split it into two disjoint open sets results in an open set that is not entirely contained within the interval.

(2) **Real Line  $\mathbb{R}$ :** The entire real line  $\mathbb{R}$  is a connected metric space. There are no disjoint open sets whose union covers the entire real line.

(3) **Open Interval in  $\mathbb{R}$ :** The open interval  $(a, b)$  in the real line  $\mathbb{R}$  is a connected metric space. It is not possible to split it into two disjoint open sets whose union forms the entire interval.

(4) Euclidean space  $\mathbb{R}^n$

(5) Circle in  $\mathbb{R}^2$

(6) Closed unit ball in  $\mathbb{R}^n$

**Q.7. Prove that the open interval  $(a, b)$  in  $\mathbb{R}$  is connected.**

**Ans.** Assume, for contradiction that  $(a, b)$  is not connected. Then, there exists disjoint open sets  $U$  and  $V$  in  $\mathbb{R}$  such that  $(a, b) \subset U \cup V$  without loss of generality, assume that  $a \in U$  &  $b \in V$ . This leads to a contradiction, thus proving that  $(a, b)$  is connected.

**Q. 2.** Let  $V$  be the vector space of all  $2 \times 2$  matrices over the field  $\mathbb{R}$  of real numbers

Let (i)  $W_1 = \{A \in V : A^2 = A\}$ ,

(ii)  $W_2 = \{A \in V : \det A = 0\}$

Show that  $W_1$  &  $W_2$  are not subspace of  $V$ .

**Ans. (i)** We know that

$$I^2 = I \quad \text{where} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in W$$

For,  $2 \in \mathbb{R}, 2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

&  $(2I)^2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

Since  $(2I)^2 \neq 2I, 2I \notin W_1$

Hence  $W_1$  is not a subspace of  $V$ .

(ii) Let  $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ .

Then  $A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

We see that let  $A = 0$  & let  $B = 0$

i.e.,  $A \in W_2$  &  $B \in W_2$

Since let  $(A + B) = I \neq 0$

$\Rightarrow A + B \notin W_2$

Hence,  $W_2$  is not a subspace of  $V$ .

**Q. 3.** Show that the intersection of two subspaces of a vector space  $V(F)$  is a subspace of  $V(F)$ .

**Ans.** Let  $W_1$  &  $W_2$  be two subspaces of a vector space  $V$ .

since  $0 \in W_1$  &  $0 \in W_2$

$\Rightarrow 0 \in W_1 \cap W_2$

and so  $W_1 \cap W_2$  is non empty.

Let  $a, b \in W_1 \cap W_2$  &  $\alpha, \beta \in F$ . Then  $a, b \in W_1$  &  $a, b \in W_2$

Since  $W_1$  is a subspace of  $V$ , so  $\alpha a + \beta b \in W_1$ .

Since  $W_2$  is a subspace of  $V$ , so  $\alpha a + \beta b \in W_2$

$\therefore \alpha a + \beta b \in W_1 \cap W_2$  for  $a, b \in W_1 \cap W_2$  and  $\alpha, \beta \in F$ .

Hence  $W_1 \cap W_2$  is a subspace of  $V(F)$ .

Since  $W_2$  is a subspace of  $V$ , so  $\alpha a + \beta b \in W_2$

$\therefore \alpha a + \beta b \in W_1 \cap W_2$  for  $a, b \in W_1 \cap W_2$  and  $\alpha, \beta \in F$ .

Hence  $W_1 \cap W_2$  is a subspace of  $V(F)$ .

**Q.4.** If  $W_1$  &  $W_2$  are two subspaces of a vector space  $V(F)$ . Then show that their sum  $W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$  is a subspace of  $V$ .



**Ans.** Clearly,  $0 = 0 + 0 \in W_1 + W_2$  so  $W_1 + W_2$  is non empty.

Let  $a, b \in W_1 + W_2$  so that

$$a = w_1 + w_2 \text{ \& } b = w'_1 + w'_2$$

where  $w_1, w'_1 \in W_1$  &  $w_2, w'_2 \in W_2$

Let  $\alpha, \beta \in F$ . Then

$$\alpha a + \beta b = \alpha (w_1 + w_2) + \beta (w'_1 + w'_2)$$

$$\alpha a + \beta b = \alpha (w_1 + w_2) + \beta (w'_1 + w'_2)$$

$$= (\alpha w_1 + \beta w'_1) + (\alpha w_2 + \beta w'_2)$$

Since  $W_1$  &  $W_2$  are subspaces of  $V$ .

$$\alpha w_1 + \beta w'_1 \in W_1 \text{ \& } \alpha w_2 + \beta w'_2 \in W_2$$

$$\Rightarrow \alpha a + \beta b \in W_1 + W_2$$

Hence  $w_1 + w_2$  is a subspace of  $V$ .

**Q. 5. Show that the set**

**$Q(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in Q\}$  is a vector space over  $Q$  with respect to the compositions:**

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2}$$

$$\alpha(a + b\sqrt{2}) = \alpha a + \alpha b\sqrt{2}, \alpha \in Q$$

**Ans.** We know  $\{Q, \sqrt{2}, +\}$  is an abelian group.

Properties of scalar multiplication.

Let  $\alpha, \beta \in Q$  &  $u = a + b\sqrt{2}, v = c + d\sqrt{2}, u, v \in Q(\sqrt{2})$

$$\alpha u = \alpha a + \alpha b\sqrt{2}, = c + d\sqrt{2}, u, v \in Q\sqrt{2}$$

$$\alpha u = \alpha a + \alpha b\sqrt{2} \in Q(\sqrt{2}).$$

$$\begin{aligned} 1. \quad \alpha(u + v) &= \alpha \{ (a + c) + (b + d)\sqrt{2} \} \\ &= \{ \alpha(a + c) + \alpha(b + d)\sqrt{2} \} \\ &= \alpha a + \alpha c + \alpha b\sqrt{2} + \alpha d\sqrt{2} \\ &= \alpha a + \alpha b\sqrt{2} + \alpha c + \alpha d\sqrt{2} \\ &= \alpha(a + b\sqrt{2}) + \alpha(c + d\sqrt{2}) \\ &= \alpha u + \alpha v. \end{aligned}$$

$$\begin{aligned} 2. (\alpha + \beta)u &= \{ (\alpha + \beta)a + (\alpha + \beta)b\sqrt{2} \} \\ &= \alpha(a + b\sqrt{2}) + \beta(a + b\sqrt{2}) \\ &= \alpha u + \beta u \end{aligned}$$

$$\begin{aligned} 3. \alpha(\beta u) &= \alpha(\beta a) + \alpha(\beta b\sqrt{2}) \\ &= (\alpha\beta)a + (\alpha\beta)b\sqrt{2} \\ &= \alpha\beta(a + b\sqrt{2}) \\ &= (\alpha\beta)u \end{aligned}$$

$$4. \quad 1.u = 1(a + b + \sqrt{2}) = a + b\sqrt{2} = u$$

Hence  $Q(\sqrt{2})$  is a vector space over  $Q$ .

**Q. 6.** If  $F$  is field of real numbers, prove that the vectors  $(1, 1, 0, 0)$ ,  $(0, 1, -1, 0)$  and  $(0, 0, 0, 3)$  in  $F^4$  are linearly independent over  $F$ .

**Ans.** Let  $x, y, z \in F$  be such that

$$x(1, 1, 0, 0) + y(0, 1, -1, 0) + z(0, 0, 0, 3) = (0, 0, 0, 0)$$

$$\Rightarrow (x, x+y, -y, 3z) = (0, 0, 0, 0)$$

$$\therefore x = 0, \quad x+y = 0, \quad -y = 0, \quad 3z = 0$$

$$\Rightarrow x = y = z = 0$$

Hence the given vectors are linearly independent over  $F$ .

**Q.7.** Show that

$S = \{(1, 0, -1, 0), (2, 1, 3, 0), (-1, 0, 0, 0), (1, 0, 1, 0)\}$  is a linearly dependent set in  $\mathbb{R}^4$ .

**Ans.** Let  $x, y, z, w \in \mathbb{R}$  be such that

$$x(1, 0, -1, 0) + y(2, 1, 3, 0) + z(-1, 0, 0, 0) + w(1, 0, 1, 0) = (0, 0, 0, 0)$$

This gives us the following equations.

$$x + 2y - z + w = 0$$

$$0x + y + 0z + 0w = 0$$

$$-x + 3y + 0z + w = 0$$

$$0x + 0y + 0z + 0w = 0$$

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 5 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{by } R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{by } R_3 \rightarrow R_3 - 5R_2$$

$$\therefore x + 2y - z + w = 0, \quad y = 0, \quad -z + 2w = 0$$

After solving above equations, we get

$$x = 1, \quad y = 0, \quad z = 2, \quad w = 1$$

Hence  $S$  is a linearly dependent set.

**Q.8.** If the vector  $(3, -1, 0, 1)$  in the subspace of  $\mathbb{R}^4$  spanned by the vectors  $\alpha_1 = (2, -1, 3, 2)$ ,  $\alpha_2 = (-1, 1, 1, -3)$  and  $\alpha_3 = (1, 1, 9, -5)$ ?

**Ans.** Suppose  $(3, -1, 0, 1)$  is spanned by  $\alpha_1, \alpha_2$  &  $\alpha_3$ . Then

$$(3, -1, 0, 1) = x(2, -1, 3, 2) + y(-1, 1, 1, -3) + z(1, 1, 9, -5); \quad x, y, z \in \mathbb{R}$$

$$2x - y + z = 3$$

$$-x + y + z = -1$$

$$3x + y + 9z = 0$$

...(1)

...(2)

...(3)



$$2x - 3y - 5z = -1 \quad \dots(4)$$

$$\text{From (1) \& (2), } x + 2z = 2 \quad \dots(5)$$

$$\text{From (3) \& (4), } x + 2z = -1 \quad \dots(6)$$

From (5) & (6), we see that the given system of equations has no solution. Hence  $(3, -1, 0, -1)$  is not in the subspace spanned by the vectors  $\alpha_1, \alpha_2$ , &  $\alpha_3$ .

**Q. 9. Let  $V = \mathbb{R}^3(\mathbb{R})$  &**

$$S = \{\alpha_1 = (1, 1, 0), \alpha_2 = (0, -1, 1), \alpha_3 = (1, 0, -1)\}$$

**Prove that  $(a, b, c) \in L(S)$  iff  $a = b + c$**

**Ans.** By definition of  $L(S)$ ,  $(a, b, c) \in L(S)$

$$\Leftrightarrow (a, b, c) = \alpha(1, 1, 0) + \beta(0, -1, 1) + \gamma(1, 0, -1); \alpha, \beta, \gamma \in \mathbb{R}$$

$$\Leftrightarrow (a, b, c) = (\alpha + \gamma, \alpha - \beta, \beta + \gamma)$$

$$\Leftrightarrow a = \alpha + \gamma, b = \alpha - \beta, c = \beta + \gamma$$

$$\Leftrightarrow a = b + c$$

**Q.10. (i) Let  $S = \{(1, 0), (0, 1)\}$  be a subset of  $\mathbb{R}^2(\mathbb{R})$ . Show that  $(3, 5)$  belongs to  $L(S)$**

**(ii) Let  $S = \{(1, 0, 0), (0, 1, 0)\} \subseteq \mathbb{R}^3(\mathbb{R})$**

**Find  $L(S)$ . Do  $(3, 2, 0)$  and  $(2, 5, 1)$  belong to  $L(S)$ ?**

**(iii) Let  $S = \{(2, 3), (1, 4)\} \subseteq \mathbb{R}^2(\mathbb{R})$ .**

**Show that  $(4, 1)$  belong to  $L(S)$ .**

**Ans.** (i) We observe that  $(3, 5) = 3(1, 0) + 5(0, 1)$  and so  $(3, 5) \in L(S)$

$$(ii) L(S) = \{\alpha(1, 0, 0) + \beta(0, 1, 0) : \alpha, \beta \in \mathbb{R}\}$$

$$= \{(\alpha + 0, 0 + \beta, 0 + 0) : \alpha, \beta \in \mathbb{R}\}$$

$$= \{(\alpha, \beta, 0) : \alpha, \beta \in \mathbb{R}\}$$

So  $(3, 2, 0) \in L(S)$  &  $(2, 5, 1) \notin L(S)$

$$(iii) \text{ Let } (4, 1) = \alpha(2, 3) + \beta(1, 4); \alpha, \beta \in \mathbb{R}$$

$$\text{or } 2\alpha + \beta = 4 \text{ \& } 3\alpha + 4\beta = 1$$

The above equations are satisfied for  $\alpha = 3, \beta = -2$ .

$$\text{Hence } (4, 1) = 3(2, 3) - 2(1, 4) \in L(S)$$

**Q. 11. Consider a matrix  $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$**

**(i) Find the eigenvalues.**

**(ii) Find the eigen vectors.**

**(iii) Demonstrate its diagonalization.**

**Ans.** (i) Characteristic equation of  $A$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix} = 0$$

$$(4-\lambda)(3-\lambda) - 2 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 10 = 0$$

After solving, we get  $\lambda_1 = 2$  &  $\lambda_2 = 5$ .

(ii) Eigenvector  $V_1 = \begin{bmatrix} x \\ y \end{bmatrix}$  corresponding to  $\lambda_1 = 2$

$$(A - 2I)V_1 = 0$$

$$\begin{bmatrix} 4-2 & 1 \\ 2 & 3-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

after solving this system of equations, we get  $V_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

For  $\lambda = 5$

$$(A - 5I)V_2 = 0$$

$$\begin{bmatrix} 4-5 & 1 \\ 2 & 3-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

after solving this system of equations we get  $V_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(iii) To diagonalize A we form the matrix P with the eigenvectors as columns and D as the diagonal matrix of eigenvalues. Therefore

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \text{ \& } D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

Then the diagonalization is given by  $A = PDP^{-1}$

$$PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A$$

This demonstrates the diagonalization of the matrix A.